A degree-two family of permutations is covered by few cosets of its certificate size

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 608-616 and its proof environment opens at line 617; the pairing is by adjacency and the retained proof range is the whole environment.

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Editorial scope. One original result of the pinned author manuscript, counted once: the coset covering bound for degree-two families of permutations (author label lem:main-ub-sym).

Allowed standard prerequisites (editorial).

- Coset covering. The family is contained in the union of T cosets, each determined by prescribing C(f) images.. It is the conclusion.
- Representative lemma. Two members with equal value admit representatives meeting exactly in the intersection of their certificates.. It is used to place members in cosets.
- Structure of a covered family. An m-covered family gives every member a certificate of size m plus one.. It bounds the number of cosets needed.

Lemma 0.1. *Let $f\colon S_n \to \{0,1\}$ be the characteristic function of a 2-intersecting family, and suppose that $C(f) \leq n-2$. If $f(\alpha) = f(\beta) = 1$ and C_α, C_β are minimum certificates for α, β (respectively), then $|C_\alpha \cap C_\beta| \geq 2$.*

Lemma 0.2. *Let f be the characteristic function of a 2-intersecting family $\mathcal{F} \subseteq S_n$, and suppose that $C(f) \leq n-2$.*

If $\alpha \in \mathcal{F}$ then $\mathcal{F}$ is m-covered for $m = C(f,\alpha) - 1$.

Lemma 0.3. *Let $\mathcal{F} \subseteq S_n$ be an m-covered family, say $\mathcal{F} \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$. Suppose that its characteristic function f has degree at most 2.*

Every α has a certificate of the form $\{(i_1, \alpha(i_1)), \ldots, (i_m, \alpha(i_m)), (i,j)\}$, and in particular, $C(f) \leq m+1$.

Lemma 0.4. *Let $\mathcal{F} \subseteq S_n$ be a 2-intersecting family whose characteristic function f has degree at most 2. Suppose that $C(f) \leq n-2$.*

The family $\mathcal{F}$ is contained in the union of at most T many $C(f)$-cosets, where

$$T = \frac{2\lfloor C(f)/2 \rfloor (C(f)-1)!}{2^{\lfloor C(f)/2 \rfloor}}.$$

Moreover, $C(f) \geq 2$.

Proof. If $\mathcal{F}$ is empty then the lemma is vacuously true. Otherwise, choose an arbitrary $\beta \in \mathcal{F}$. Theorem 0.1, applied to $\alpha = \beta$, shows that $C(f,\beta) \geq 2$. Theorem 0.2 shows that $\mathcal{F}$ is m-covered, where $m = C(f,\beta) - 1$. Without loss of generality, we can assume that $\mathcal{F} \subseteq [1 \mapsto 1] \cup \cdots \cup [m \mapsto m]$.

Theorem 0.3 shows that $C(f) \leq m+1$, and so $m = C(f) - 1$ and $C(f) = C(f,\beta) \geq 2$. The theorem also shows that every $\alpha \in \mathcal{F}$ has a certificate C_α of the form $\{(1,\alpha(1)), \ldots, (m,\alpha(m)), (i,j)\}$. It also implies that $\mathcal{F}$ is not $(m-1)$-covered (otherwise its certificate complexity would be at most $m < C(f)$). Therefore, if X is any set of at most $m-1$ compatible cosets, then there is $\alpha_X \in \mathcal{F}$ which is not contained in the union of the cosets in X. Fix a certificate C_X for α_X of the form $\{(1,\alpha_X(1)), \ldots, (m,\alpha_X(m)), (i_X, j_X)\}$.

We think of the certificates C_α, C_X as colored using $m+1$ colors: for $k \in [m]$, a pair $(k,*)$ is colored k, and the remaining pair (i,j) is colored $m+1$. We can determine the color of a pair (i',j') given only the pair: if $i' \in [m]$ then the color is i', and otherwise the color is $m+1$.

We will prove the lemma by showing that if $\alpha \in \mathcal{F}$ then C_α is one of at most T possible certificates. Since C_α corresponds to a $C(f)$-coset containing α, the lemma would follow. In the proof, we will freely identify a pair (i,j) with the corresponding coset $[i \mapsto j]$, and from now on we will only ever mention cosets.

The proof is slightly different depending on the parity of $C(f)$. We start with the case in which $C(f) = 2r$ is even. Let $\alpha \in \mathcal{F}$. We define a sequence $X_0 \subset \cdots \subset X_r \subseteq C_\alpha$, where X_s is a collection of $2s$ compatible cosets, as follows. The starting point is $X_0 = \emptyset$. Now let $s < r$, and suppose that X_s has been defined. Since $|X_s| = 2s \leq C(f) - 2 = m - 1$, there is a permutation $\alpha_{X_s} \in \mathcal{F}$ which lies outside of the union of the cosets in X_s. According to Theorem 0.1, the certificates C_α and C_{X_s} must have at least two cosets in common q_1, q_2. The choice of α_{X_s} guarantees that $q_1, q_2 \notin X_s$, and we define $X_{s+1} = X_s \cup \{q_1, q_2\}$. By construction, $X_{s+1} \subseteq C_\alpha$, and so the cosets in X_{s+1} are indeed compatible.

Note that $|X_r| = 2r = |C_\alpha|$, and so $X_r = C_\alpha$. For $s < r$, given X_s, the set $X_{s+1} \setminus X_s$ consists of two cosets belonging to C_{X_s}. Moreover, since $X_{s+1} \subseteq C_\alpha$ and C_α contains one coset of each color, the colors of the two cosets in $X_{s+1} \setminus X_s$ are different from the colors of the cosets in X_s. Since C_{X_s} also contains one coset of each color, this means that there are $\binom{|C_{X_s}| - |X_s|}{2} = \binom{C(f)-2s}{2}$ choices

for $X_{s+1} \setminus X_s$. In total, the number of possible choices for C_α is

$$\prod_{s=0}^{r-1} \binom{C(f)-2s}{2} = \frac{C(f)!}{2^r},$$

matching the formula for T in the statement of the lemma.

Now suppose that $C(f) = 2r+1$ is odd. Let $\alpha \in \mathcal{F}$. We define a sequence $X_0 \subset \cdots \subset X_r \subseteq C_\alpha$, where X_s is a collection of $2s+1$ compatible cosets, as follows. We start by defining X_0. According to Theorem 0.1, the certificates C_α and $C_\emptyset$ must contain at least two cosets in common q_1, q_2. At least one of these cosets is of the form $(k, \alpha(k))$ for some $k \in [m]$. We define $X_0 = \{(k, \alpha(k))\}$. Now let $s < r$, and suppose that X_s has been defined. Since $|X_s| = 2s+1 \leq C(f) - 2 = m-1$, we can define X_{s+1} as in the case of even $C(f)$.

Note that $|X_r| = 2r+1 = |C_\alpha|$, and so $X_r = C_\alpha$. The certificate X_0 contains a single coset $(k, \ell) \in C_\emptyset$, where $k \in [m]$, and so there are $m = C(f) - 1$ choices for X_0. For $s < r$, given X_s, there are $\binom{|C_{X_s}|-|X_s|}{2} = \binom{C(f)-2s-1}{2}$ choices for X_{s+1} (this is the same as in the case of even $C(f)$). In total, the number of possible choices for C_α is

$$(C(f)-1) \prod_{s=0}^{r-1} \binom{C(f)-2s-1}{2} = (C(f)-1)\frac{(C(f)-1)!}{2^r},$$

matching the formula for T in the statement of the lemma. □

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